

FLUID MECHANICS I
PROBLEM 1 (10 POINTS).

28-10-2018
Time: 90 minutes

NAME:.....

GRADE:.....

Figure (a) below shows a cylinder of radius a_0 closed at the bottom that contains a volume $\pi a_0^2 h_0$ of water. The water is in contact with the stagnant atmosphere at a pressure p_a .

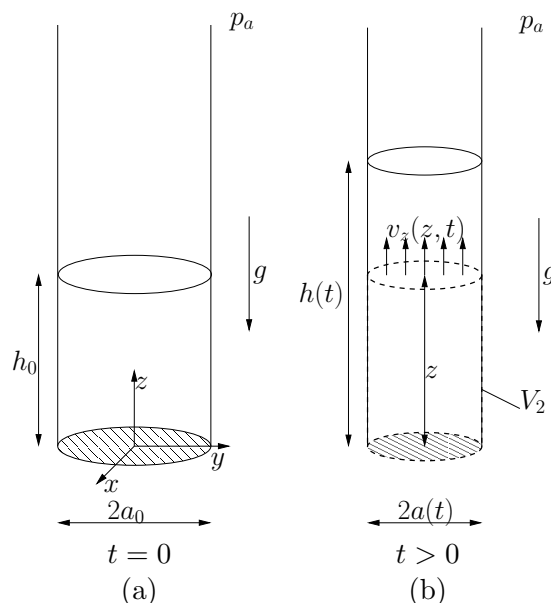
At time $t = 0$ the cylinder radius $a(t)$ starts decreasing and the level of water rises, as shown in figure (b) for a generic instant of time t . If $a(t)$ is known,

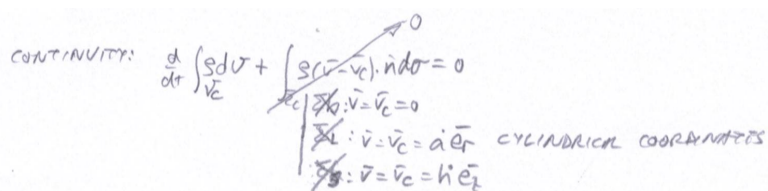
1. Define **clearly** a **control volume** and use the mass conservation principle to determine the level of water $h(t)$ and the velocity of the free surface $\dot{h}(t)$ for a generic instant of time t .
2. Using **control volume** V_2 , as shown in the figure (b), use the continuity equation to obtain the z-component of the velocity $v_z(z, t)$ in the liquid at an arbitrary level z , as sketched in figure b.

Using the previous results, and specifying the control volume used in the analysis,

3. Determine separately the force exerted by both the liquid and the atmosphere on the cylinder. Calculate the **net force** exerted by both fluids on the cylinder.
4. Using the **kinetic energy equation**, and neglecting v_r^2 in the kinetic energy, obtain the mechanical power exerted on the solid walls by both the liquid and the atmosphere. Determine the **net power** exerted by both fluids on the liquid.

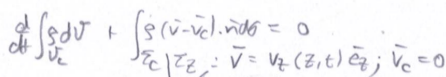
Note: Neglect viscous dissipation within the liquid, viscous stresses at the free surface and take the pressure in the air as a constant p_a





THEREFORE: $\frac{d}{dt} [pV_e] = 0 \rightarrow \rho \pi a^2 (H) \cdot h(H) = \rho \pi a_0^2 h_0$

$$\boxed{h(t) = \frac{a_0^2}{a^2(t)} h_0} \quad (1) \Rightarrow \boxed{\dot{h} = -\frac{2a_0^2}{a^3} \dot{a} h_0 = -2h(t) \frac{\dot{a}(t)}{a(t)}} \quad (2)$$



$$\frac{d}{dt} [s v_z] + s v_z \pi a^2 = 0 \rightarrow \frac{d}{dt} (a^2 z) + \frac{1}{2} a^2 = 0$$

$$V_z = -\frac{1}{a^2} \frac{da^2}{dt} Z = \left[-\frac{2\dot{a}(t)}{a(t)} \right] Z = V_z(Z, t) \rightarrow \text{For } Z=h: V_z = -\frac{2\dot{a}}{a} h = \dot{h}$$

USING C.V. SPECIFIED IN #1:

MOMENTUM-Z: $\frac{d}{dt} \int_{V_c} \rho v_z dv + \int_{S_c} \rho v_z \vec{n} \cdot d\vec{s} = \int_{V_c} -\rho \ddot{x}_0 dz + \int_{S_c} \bar{\tau}_{xz} \cdot n_x ds + \int_{V_c} f_z dv$

$\begin{array}{|c|c|} \hline \bar{\tau}_{xz}|_{z=L} & \bar{\tau}_{xz}|_{z=b} \\ \hline \end{array}$

NEGLIGIBLE

$\boxed{\begin{array}{|c|c|} \hline \bar{\tau}_{xz}|_{z=L} & \bar{\tau}_{xz}|_{z=b} \\ \hline \end{array}} = F_{w \rightarrow L} = -F_{L \rightarrow w}$

DIRECTION Z: $\frac{d}{dt} \int_{V_c} \rho V_z dV = -F_{L \rightarrow W} - P_a \cdot \pi a^2 - \rho g V_c$

$$F_{L \rightarrow W} = - \frac{d}{dt} \int_V \rho v_z(z,t) dV + P_a \cdot \pi a^2 - \rho g V_c = - \frac{d}{dt} \int_0^{h(t)} \rho \left(-z \frac{a}{\alpha} \right) \pi a^2 dz - P_a \pi a^2 - \rho g V_c$$

$$F_{L \rightarrow W} = \frac{d}{dt} \left[\frac{1}{2} \pi a \dot{h}^2 \right] - R \pi a^2 - \rho g V_L = \pi \left[\dot{a}^2 h^2 + a \ddot{a} h^2 + 2 a \dot{a} h \dot{h} \right] - R \pi a^2 - \rho g V_L$$

$$\text{USING (1) (2):}$$

$$= 5\pi \left[\dot{a}^2 \frac{a_0^4}{a^4} h_0^2 + a \ddot{a} \frac{a_0^4}{a^4} h_0^2 + 2a \dot{a} \frac{a_0^2}{a^2} h_0 \cdot 2 \frac{\dot{a}}{a} \cdot \frac{a_0^2}{a^2} h_0 \right] - P_a \pi a^2 - \rho g \pi a_0^2 h_0$$

$$F_{L \rightarrow W} = 8\pi \frac{a_0^4}{a^4} h_0^2 [a\ddot{a} - 3\dot{a}^2] - P_a \pi a^2 - 8g\pi a_0^2 h_0$$

$$F_{A \rightarrow W} = \rho_a \pi a^2$$

$$F = F_{L \rightarrow W} + F_{A \rightarrow W} = 8\pi a_0^2 \left[\frac{h^2}{a^4} [a\ddot{a} - 3\dot{a}^2] - g h_0 \right]$$

④ POWER: SAME CONTROL VOLUME AS IN #1

$$\frac{d}{dt} \int_{V_C} \rho \mathbf{v}^2 dV + \int_{V_C} \rho \mathbf{v}^2 (\mathbf{v} - \mathbf{v}_i) \cdot \mathbf{n} d\mathbf{r} = \int_{V_C} -p \mathbf{\bar{v}} \cdot \mathbf{n} d\mathbf{r} + \int_{V_C} \mathbf{\bar{v}} \cdot \mathbf{\bar{e}} \cdot \mathbf{n} d\mathbf{r} + \int_{V_C} \mathbf{\bar{F}}_{\text{int}} \cdot \mathbf{v} dV - \left[\int_{V_C} \rho \mathbf{v} \cdot \mathbf{v} dV + \int_{V_C} \rho \mathbf{v}^2 dV \right]$$

$$\frac{d}{dt} \int_{V_C} \rho \frac{v_z^2(z,t)}{2} dV = -\dot{W}_{L \rightarrow W} - P_a \pi a^2 \dot{h} - \int_{V_C} \rho g v_z(z,t) dV$$

$$\dot{W}_{L \rightarrow W} = -\frac{d}{dt} \int_{V_C} \rho \frac{v_z^2(z,t)}{2} dV - P_a \pi a^2 \dot{h} - \int_{V_C} \rho g v_z(z,t) dV =$$

$$= -\frac{d}{dt} \left[\frac{\rho \pi a^2}{2} \int_0^h v_z^2(z,t) dz \right] - P_a \pi a^2 \dot{h} - \int_0^h \rho g v_z(z,t) \pi a^2 dz =$$

$$= -\frac{d}{dt} \left[\frac{\rho \pi a^2}{2} \frac{4 \dot{a}^2}{a^2} \frac{h^3}{3} \right] - P_a \pi a^2 \dot{h} + \rho g \pi a^2 \frac{\dot{a}}{a} \frac{h^2}{2}$$

$$= -\frac{2}{3} \rho \pi \frac{d}{dt} [\dot{a}^2 h^3] - P_a \pi a^2 \dot{h} + \rho g \pi a \dot{a} h^2$$

$$= -\frac{2}{3} \rho \pi \left[2 \dot{a} \ddot{a} \frac{a_0^6}{a^6} h_0^3 + \dot{a}^2 \cdot 3 h^2 \dot{h} \right] - P_a \pi a^2 \dot{h} + \rho g \pi a \dot{a} h^2$$

$$= -\frac{2}{3} \rho \pi \left[2 \dot{a} \ddot{a} \frac{a_0^6}{a^6} h_0^3 + \dot{a}^2 \cdot 3 \cdot \frac{a_0^4}{a^4} h_0^2 \cdot 2 h \frac{\dot{a}}{a} \frac{a_0^2}{a^2} h_0 \right] - P_a \pi a^2 \dot{h} + \rho g \pi a \dot{a} h^2$$

$$= -\frac{4}{3} \rho \pi \left(\frac{a_0}{a} \right)^6 h_0^3 \dot{a} \left[\ddot{a} - 3 \frac{\dot{a}^2}{a} \right] + P_a \pi a^2 \frac{2 a_0^2}{a} \dot{a} h_0 + \rho g \pi a \dot{a} h^2$$

$$\dot{W}_{L \rightarrow W} = -\frac{4}{3} \rho \pi \left(\frac{a_0}{a} \right)^6 h_0^3 \dot{a} \left[\ddot{a} - 3 \frac{\dot{a}^2}{a} \right] + P_a \pi a^2 \frac{2 a_0^2}{a} \dot{a} h_0 + \rho g \pi a \dot{a} h^2$$

$$\dot{W}_{A \rightarrow W} = \int_{V_C} -p \mathbf{\bar{v}} \cdot \mathbf{n} d\mathbf{r} + \int_{V_C} \mathbf{\bar{v}} \cdot \mathbf{\bar{e}} \cdot \mathbf{n} d\mathbf{r} = -P_a \dot{a} 2 \pi a h = -P_a \pi 2 \frac{a_0^2}{a} \dot{a} h_0$$

$$\dot{W} = \dot{W}_{L \rightarrow W} + \dot{W}_{A \rightarrow W} = -\frac{4}{3} \rho \pi \left(\frac{a_0}{a} \right)^6 h_0^3 \dot{a} \left[\ddot{a} - 3 \frac{\dot{a}^2}{a} \right] + \rho g \pi a \dot{a} h^2$$